

Review:

Boltzmann probability:

$$S = -k \sum p_i \ln p_i$$

Entropy of mixing:

$$S = R \sum x_i \ln x_i$$

Standard Entropy:

$$S^\circ = \int_0^{298} \frac{C_p}{T} dT$$

1 Section 4A - Gibbs and Auxillary functions

Gibbs:

$$dG = dH - Tds$$

1.1 Gibbs derivation

All systems tend toward a lower "Energy state". This process acquires entropy (maximizes configuration). Consider the three laws applied to a reversible process:

- 1st law:
 $dU = \delta q - \delta w$
 $dU = \delta q_{rev} - PdV$ (I)
- 2nd law:
 $dS \geq \frac{\delta q}{T}$
 $dS = \frac{\delta q_{rev}}{T}$
 $\delta q_{rev} = Tds$ (II)
- 3rd law:
 $S = 0$ at 0K
- Enthalpy:
 $H \equiv U + PV$
 $dH = dU + PdV + VdP$ (III)

1.2 cont.

$$(II) \circ (I): dU = Tds - PdV \text{ (IV)}$$

$$(IV) \circ (III): dH = (Tds - PdV) + PdV + VdP$$
$$dH = TdS + VdP \text{ (IV)}$$

1.3 For a process at constant T and P

$$(IV) \Big|_{P,T}: dH = Tds$$

- This equation is a criteria for sponteneity
- It displays an equilibrium which can be used to predict the direction of a reaction
- This equation balances both:
 - enthalpy (naturally lower energy state is preferred)
 - entropy (naturally tends towards a statistically favored state)

1.4 Gibbs Free Energy

$$G \equiv H - TS$$

$$\Delta G = \Delta H - T\Delta S$$

$$dG = dU + VdP + PdV - SdT - TdS$$

$$dG = \delta q - \delta w + VdP + PdV - SdT - TdS$$

$$dG = TdS - PdV + VdP + PdV - SdT - TdS$$

$$dG = VdP - SdT$$

The final line indicates that the equilibrium criteria varies with temperature and pressure

$\left.\frac{\delta G}{\delta T}\right _P = -S$	$\left.\frac{\delta G}{\delta P}\right _T = V$	$\left.\frac{\delta H}{\delta P}\right _S = V$	$\left.\frac{\delta H}{\delta S}\right _P = T$
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1.5 Gibbs: Standard State

$$\Delta G^\circ = \Delta H^\circ - \Delta S^\circ T$$

2 Section 4B

Objectives:

1. Identify and appreciate other thermodynamic ensembles and their Auxiliary functions
2. Define thermodynamic parameters based on partial derivatives of their associated ensembles
3. Outline experiments in different ensembles
4. Discuss and utilize the Legendre transform to convert between Auxiliary functions

2.1 First order Homogenous functions

$$F(x_1, \dots, x_i) = \frac{dF}{dx_1}x_1 + \frac{dF}{dx_2}x_2 + \dots + \frac{dF}{dx_i}x_i$$

Applying this theorem to Thermodynamic equations:

$$G(P, T, n) = \frac{dG}{dP}P + \frac{dG}{dT}T + \frac{dG}{dn}n$$

$$dG(P, T, n) = \frac{dG}{dP}dP + \frac{dG}{dT}dT + \frac{dG}{dn}dn$$

2.2 Auxillary functions(Ensembles)

Ensemble	Potential
Micro-canonical	$U(S, V, N)$ $dU = TdS - PdV + \mu dn$ $T = \frac{\partial U}{\partial S}, \quad P = -\frac{\partial U}{\partial V}, \quad \mu = \frac{\partial U}{\partial N}$
Canonical	$F(T, V, N) = U - TS$ $dF = -SdT - PdV + \mu dn$ $S = -\frac{\partial F}{\partial T}$
Enthalpic	$H(S, P, N) = U + PV$ $dH = TdS - VdP + \mu dn$ $V = \frac{\partial H}{\partial P}$
Gibbsian	$G(T, P, N) = H - TS$ $dG = -SdT + VdP + \mu dn$ $S = -\frac{\partial G}{\partial T}$
Grand-canonical	$K(T, V, \mu) = F - \mu n$ $dK = -SdT + VdP - nd\mu$ $n = \frac{\partial K}{\partial \mu}$

2.2.1 Micro-canonical ensemble

$$U(S, V, n) = \frac{dU}{dS}S + \frac{dU}{dV}V + \frac{dU}{dn}n$$

Where:

$$\frac{dU}{dS} = T, \quad \frac{dU}{dV} = -P, \quad \frac{dU}{dn} = \mu$$

2.2.2 Canonical ensemble

$$F(T, V, n) = U - TS = \frac{dF}{dT}T + \frac{dF}{dV}V + \frac{dF}{dn}n$$

Where:

$$\frac{dF}{dT} = -S, \quad \frac{dF}{dV} = -P, \quad \frac{dF}{dn} = \mu$$

2.2.3 Enthalpic ensemble

$$H(S, P, n) = U + PV = \frac{dH}{dS}S + \frac{dH}{dP}P + \frac{dH}{dn}n$$

Where:

$$\frac{dH}{dS} = T, \quad \frac{dH}{dP} = V, \quad \frac{dH}{dn} = \mu$$

2.2.4 Gibbsian ensemble

$$G(T, P, n) = H - TS = \frac{dG}{dT}T + \frac{dG}{dP}P + \frac{dG}{dn}n$$

Where:

$$\frac{dG}{dT} = -S, \quad \frac{dG}{dP} = V, \quad \frac{dG}{dn} = \mu$$

2.3 Legendre Transform

Consider:

$$f(x, y) = \frac{df}{dx}x + \frac{df}{dy}y$$

$$df = \frac{df}{dx}dx + \frac{df}{dy}dy$$